

# SlotFlow: From Toy Problems to LISA

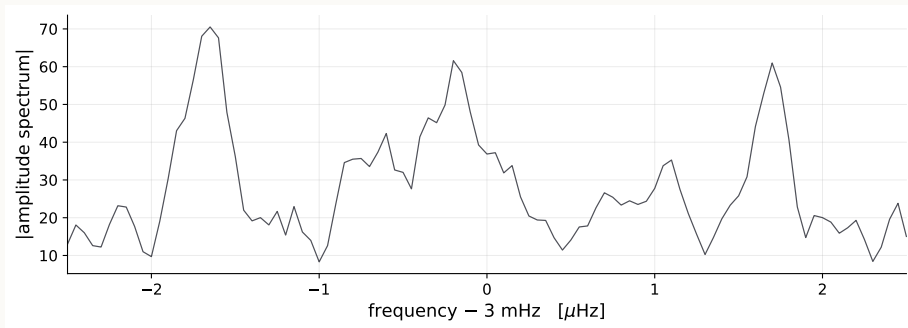
Two hands-on tutorials on amortized trans-dimensional inference

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Niklas Houba & Giovanni Giarda · ETH Zürich

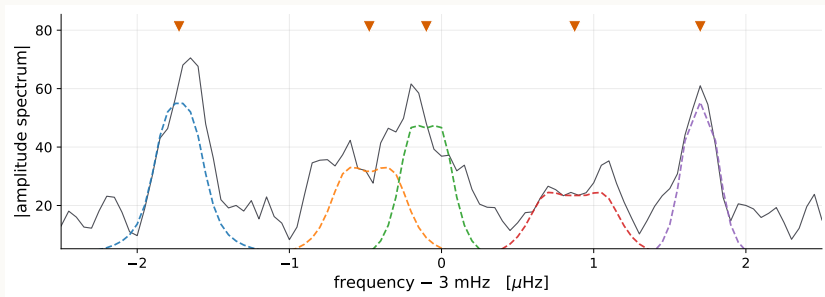
*LISA AI Hackathon: Machine Learning Methods for LISA* · ESA ESTEC

# Warm-up: how many sources are in this signal?



**Write down a number — and note how confident you are.**

# The answer: five — some only visible as overlapping plateaus



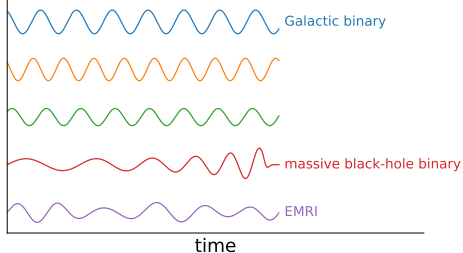
▼ source positions      dashed colored curves: the five injected sources individually

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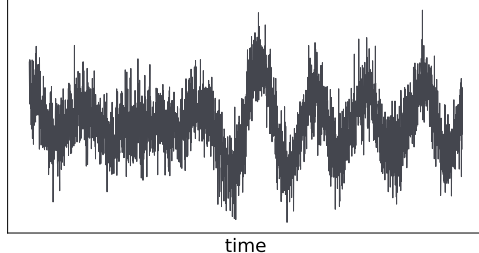
**Counting by eye already fails here — a model must infer  $K$  *and* the parameters from the mixture alone.**

# LISA is a catalogue problem: many sources, one stream

many sources, many kinds (schematic) ...



... one noisy observed data stream



$\sim 10^4$  Galactic binaries + MBHBs, EMRIs, backgrounds · no one-event-at-a-time segmentation · calibrated posteriors required

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**There is no single event to fit — the deliverable is the whole catalogue.**

# The target is an unordered set — and its size is part of the answer

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$$y = \sum_{k=1}^K s(\theta_k) + n, \quad K \text{ unknown}$$

a set whose size is itself unknown — infer  $p(K, \{\theta_1, \dots, \theta_K\} \mid y)$

## order is meaningless

“source 3” is not a thing;  
relabeling changes nothing

## overlap blurs membership

near the resolution limit,  
who-is-who gets ambiguous

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## $K$ is a parameter

the *dimension* of the  
problem is uncertain too

**Three structural difficulties — each one returns in the tutorials.**

# Sampling: costly to rerun — amortization: fixed-dimensional

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## global trans-dimensional fit

principled reference

~days of sampling,  
rerun per dataset & prior

## standard amortized posterior

milliseconds per signal

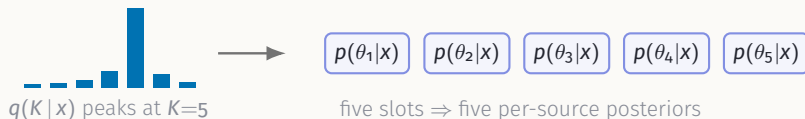
fixed dimension:  
 $K$  is a separate step

## missing capability

infer  $K$  and all  $\theta_k$   
jointly, in one pass

SlotFlow targets the third column — as a **complement** to the global fit (re-analysis, proposals, triage), not a replacement.

# SlotFlow: encode once, count, instantiate, share one flow



Only the encoder knows the physics; counting, slots, and the flow are shared. Cost grows  $\sim$ linearly with  $\hat{K}$ .

# Sets have no first element — so training needs assignment

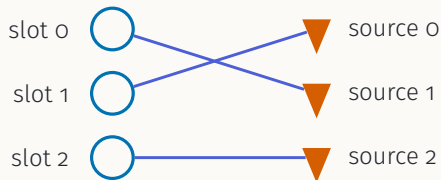
the same two-source answer, written twice

order A:  $[\theta_{\text{low}}, \theta_{\text{high}}]$

index-wise loss:  
**small** ✓

order B:  $[\theta_{\text{high}}, \theta_{\text{low}}]$

index-wise loss:  
**huge** ✗



compare every slot to every source;  
keep the best one-to-one match

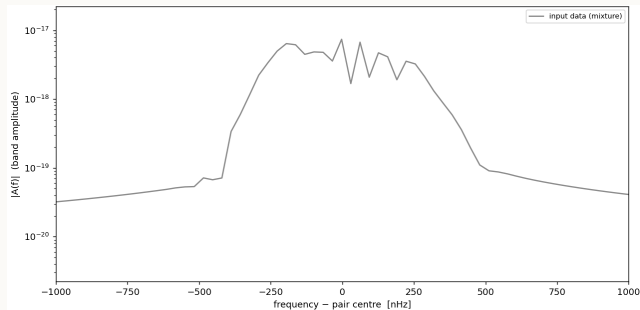
The assignment, not the slot index, defines who is compared with whom — for the **loss** and for **evaluation**.

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**Tutorial 1: break a model with the index-wise loss — on purpose — then fix it.**



# Sometimes even the number of sources is not obvious

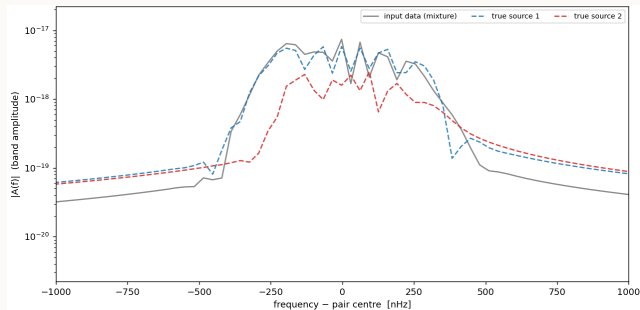


one LISA Galactic-binary band spectrum — the mixture is **all the model sees**

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**What would you report: one broad source — or two overlapping sources?**

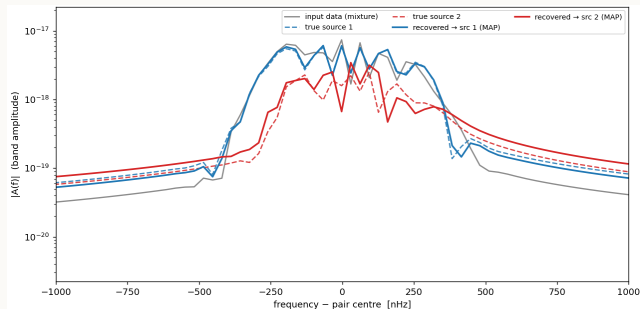
# This mixture contains two sources only 17.1 nHz apart



— observed mixture      - - / - - the two injected sources

**Two sources this close no longer look like two peaks — one source or two becomes a genuine inference problem.**

# Current SlotFlow recovers both overlapping sources



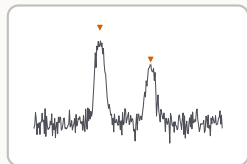
$K_{\text{true}} = 2$     $\hat{K} = 2$    dashed = truth   solid = recovered (MAP)  
**src 1:**  $\Delta f = +2.6$  nHz, sky  $1.6^\circ$    **src 2:**  $\Delta f = +5.7$  nHz, sky  $2.9^\circ$

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**One representative success in the overlapping regime — not a claim that every confused pair is solved.**

# One problem, three levels of realism

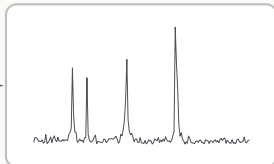
## toy sets



Tutorial 1 (this morning):  
the set mechanism

+ noise  
+ waveforms

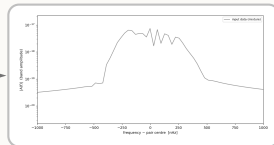
## sinusoidal mixtures



Tutorial 2:  
the pretrained model

+ real LISA  
physics

## overlapping LISA sources



current research: your warm-up  
example, the 17.1 nHz pair

Left to right the encoder and physics become more realistic — the central set-inference problem never changes.

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**The tutorials teach the basics you need to confront the full LISA problem.**

# The two tutorials form one progression

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## Tutorial 1 — build the mechanism

09:50–10:30 · toy spectral sources

- infer a deliberately simplified toy catalogue;
- deliberately **break** ordered training;
- **one coding TODO**: Hungarian matching.

## Tutorial 2 — interrogate the model

11:00–12:00 · the pretrained SlotFlow

- inspect  $q(K|x)$  and candidate slots;
- diagnose **compound failures**;
- **one coding task**: a better catalogue decision rule;
- separate detection from calibration.

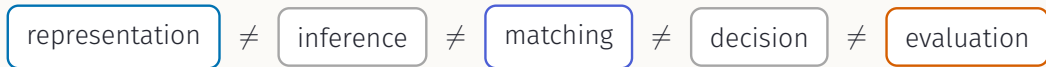
One coding task per session · CPU-only, local or Colab · executed solutions released at each session's end.

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**First you build the mechanism — then nothing about the real model is magic.**

# One discipline to carry through the morning

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Today's failures are engineered *on purpose* — the skill we practice is naming **which layer** failed, instead of letting one layer take the blame for another.

**And remember the warm-up: your number, and how unsure you were.**

By noon you will know how a model turns that into a calibrated answer — and how to check it.

## Set up now

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TODO: workshop link / QR

Colab link or the `slotflow_tutorials` folder — both work the same

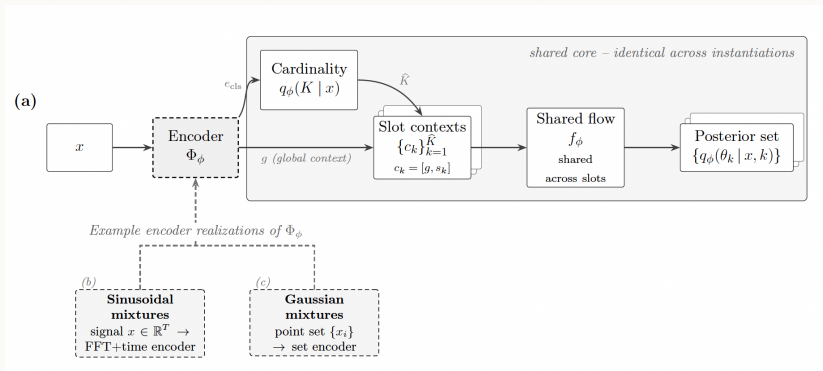
1. open `slotflow_tutorial_1_toy.ipynb`;
2. run the first cell — it fetches everything it needs;
3. no GPU, no LISA software stack.

Optional live-model checkpoint (464 MB): command in the README — coffee-break wifi, not now.

**Backup**



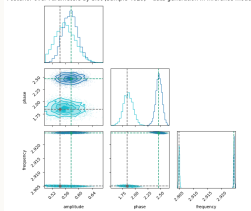
# Backup: the architecture, as published



The cardinality head  $q_\phi(K | x)$  instantiates  $\hat{K}$  slot contexts  $c_k = [g, s_k]$ ; one shared conditional flow yields the posterior set. [arXiv:2511.23228](https://arxiv.org/abs/2511.23228)

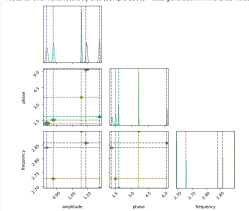
# Backup: quantitative results (held-out mixtures)

Posterior over Parameters by Slot (Sample 4029) - data generation in inference mode



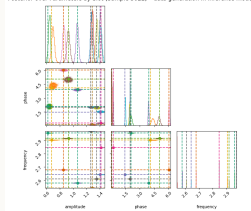
$K = 2$

Posterior over Parameters by Slot (Sample 1898) - data generation in inference mode



$K = 5$

Posterior over Parameters by Slot (Sample 3022) - data generation in inference mode



$K = 10$

- counting: **99.85%** correct over  $10^4$  held-out signals;
- vs. RJ-MCMC: amplitude/phase posteriors match (Wasserstein  $< 0.01/0.03$ ); frequency **2–3× broader**;
- calibrated (simulation-based calibration);  $\sim$ ms per signal  $\approx 10^6\times$  faster than the sampler.

## Backup: the flow, in one paragraph

- one conditional normalizing flow, **shared across slots**: alternating rational-quadratic-spline and affine autoregressive transforms (nflows),  $\sim 26\text{--}31\text{M}$  parameters;
- slot context  $\mathbf{c}_k = [\mathbf{g}, \text{onehot}(\mathbf{k})]$  — slot  $k$ 's posterior is identical no matter how many other slots exist;
- training: Hungarian-matched flow NLL — assignment on a *detached* cost matrix, loss gathered from the *live* one (Tutorial 1's TODO, with squared error in place of NLL);
- circular phase enters as  $(\cos \varphi, \sin \varphi)$ .

## Backup: how the sandbox maps to LISA (and what is missing)

|              | tutorial sandbox                               | real Galactic binaries                                                                   |
|--------------|------------------------------------------------|------------------------------------------------------------------------------------------|
| source model | sinusoid ( $\mathbf{a}, \varphi, \mathbf{f}$ ) | quasi-monochromatic GW ( $\mathcal{A}, \varphi_0, \mathbf{f}, \dot{\mathbf{f}}, \dots$ ) |
| band         | 2.5–3.0 Hz (scaled)                            | mHz                                                                                      |
| crowding     | $\Delta \mathbf{f} \gtrsim 2 \times$ Rayleigh  | confusion foreground                                                                     |
| noise        | interpolated white-ish                         | instrument + confusion PSD, TDI                                                          |
| response     | none                                           | full LISA response                                                                       |

Every omission is named on purpose:  $\dot{\mathbf{f}}$ , the response, TDI noise, and inter-source correlations are the roadmap to a real instantiation.

## Backup: references

- **SlotFlow** — Houba, Giarda, Speri, arXiv:2511.23228 · [github.com/nhouba/slotflow-inference](https://github.com/nhouba/slotflow-inference)
- **Erebor global fit** — Katz et al., PRD 111, 024060 (2025), arXiv:2405.04690
- **LISA** — Definition Study Report, arXiv:2402.07571
- amortized / simulation-based inference for GWs: DINGO-family, NPE for Galactic binaries, population-level SBI — pointers in the break.